Studying Client Heterogeneity in Federated Graph Learning for AML-like Directed Multigraphs

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1 Introduction

Federated learning is attractive for privacy-sensitive domains such as anti-money laundering (AML), where multiple institutions may wish to train models collaboratively without sharing raw data. However, standard federated learning methods can perform poorly when clients differ strongly in their local data distributions. In practice, such heterogeneity may appear not only as label imbalance, but also as differences in graph topology and in the structural patterns present in local subgraphs.

This project studies federated graph learning in an AML-like directed multigraph setting. The main goal is to systematically investigate how different forms of client heterogeneity affect model performance and convergence. Besides label skew, the project considers heterogeneity in degree distribution, graph size, and the frequency or composition of structural patterns such as cycles, scatter-gather motifs, and bipartite patterns. For example, one client may contain many cycle-based patterns, while another may contain relatively more scatter-gather or bipartite structures.

The broader aim is to understand which types of heterogeneity are most harmful for standard federated training, and whether simple model-sharing strategies are sufficient to deal with these cases.

2 Methodology

The project starts from a synthetic directed multigraph dataset for pattern-detection tasks. In the first phase, the focus will be on a standard federated learning setting in which clients only share model parameters or gradients with the server. This provides a clear baseline for studying the effect of heterogeneity.

Initially, existing partitioning methods such as METIS and Louvain will be used to create federated client splits. Afterwards, the synthetic dataset generator will be extended to create independent client/graph with controlled types and levels of heterogeneity.

For the learning models, the project will use baselines for directed multigraph pattern detection, such as Multi-GNN and MEGA-GNN, together with simple federated strategies including global federated training and possible more advanced strategies such as local-only training, cluster-based sharing, and interpolation between global and local models.

The empirical study will measure how different heterogeneity settings influence training stability, convergence behaviour, and final predictive performance. Based on these observations, the project will evaluate which simple sharing strategies are most suitable under different heterogeneous conditions.